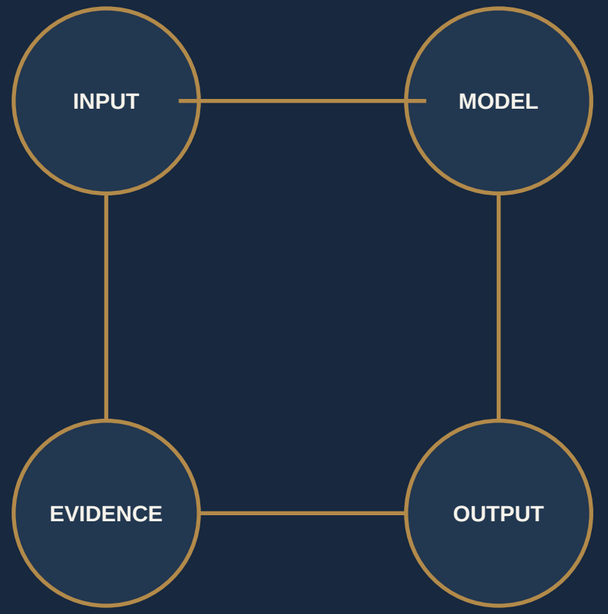
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Simulation

AI.409 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define physics simulation.
- Apply key methods including robot models.
- Build or analyze a practical system involving environments.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Physics simulation

02

METHODS

Robot models

03

SYSTEMS

Environments

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Physics simulation

Physics simulation

02 · Robot models

Robot models

03 · Environments

Environments

04 · Sensors

Sensors

05 · Actuators

Actuators

06 · Simulation control and simula

Simulation control and simulation-to-real considerations

07 · Mandatory practical: Build a s

Mandatory practical: Build a simulated robot

08 · Environment and complete auton

Environment and complete autonomous mission using an appropriate robotics simulator

Core ideas, made explicit

AI.409 • Distinguish definitions, assumptions and evidence

- Physics simulation
- Robot models
- Environments
- Sensors
- Actuators

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.409 • Distinguish definitions, assumptions and evidence

- Simulation control and simulation-to-real considerations
- Mandatory practical: Build a simulated robot
- Environment and complete autonomous mission using an appropriate robotics simulator

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Physics simulation

INPUTS
Robot models

METHOD
Environments

OUTPUT
Sensors

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving physics simulation.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using physics simulation.
- Implement or simulate a baseline using robot models.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.